

# Simultaneous Removal of Antibiotic Resistant Bacteria, Antibiotic Resistance Genes, and Micropollutants by FeS<sub>2</sub>@GO-Based Heterogeneous Photo-Fenton Process

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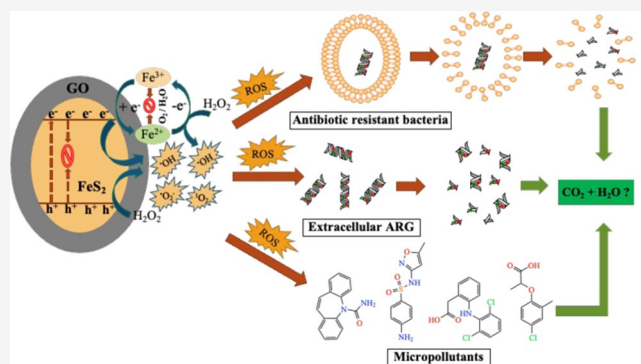
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Supporting Information

**ABSTRACT:** The co-occurrence of various chemical and biological contaminants of emerging concerns has hindered the application of water recycling. This study aims to develop a heterogeneous photo-Fenton treatment by fabricating nano pyrite (FeS<sub>2</sub>) on graphene oxide (FeS<sub>2</sub>@GO) to simultaneously remove antibiotic resistant bacteria (ARB), antibiotic resistance genes (ARGs), and micropollutants (MPs). A facile and solvothermal process was used to synthesize new pyrite-based composites. The GO coated layer forms a strong chemical bond with nano pyrite, which enables to prevent the oxidation and photocorrosion of pyrite and promote the transfer of charge carriers. Low reagent doses of FeS<sub>2</sub>@GO catalyst (0.25 mg/L) and H<sub>2</sub>O<sub>2</sub> (1.0 mM) were found to be efficient for removing 6-log of ARB and 7-log of extracellular ARG (e-ARG) after 30 and 7.5 min treatment, respectively, in synthetic wastewater. Bacterial regrowth was not observed even after a two-day incubation. Moreover, four recalcitrant MPs (sulfamethoxazole, carbamazepine, diclofenac, and mecoprop at an environmentally relevant concentration of 10 μg/L each) were completely removed after 10 min of treatment. The stable and recyclable composite generated more reactive species, including hydroxyl radicals (HO•), superoxide radicals (O<sub>2</sub><sup>•−</sup>), singlet oxygen (<sup>1</sup>O<sub>2</sub>). These findings highlight that the synthesized FeS<sub>2</sub>@GO catalyst is a promising heterogeneous photo-Fenton catalyst for the removal of emerging contaminants.

**KEYWORDS:** emerging contaminants (EC), antibiotic resistant bacteria (ARB), antibiotic resistance genes (ARGs), micropollutants (MPs), heterogeneous photo-Fenton, FeS<sub>2</sub>@GO



## INTRODUCTION

A great variety of biological and chemical emerging contaminants (EC) have been ubiquitously detected in water environments. However, all the contaminants are not easily removed by the conventional water or wastewater treatment processes, thus posing risks for public health and ecosystems. For example, biological contaminants, such as antibiotic resistant bacteria (ARB) and antibiotic resistance genes (ARGs), are widely detected in treated water, thus posing a threat to public health. As a chemical contaminant, micropollutants also have the potential to cause serious environmental and human health effects.<sup>1–3</sup>

Advanced oxidation processes (AOPs) are proposed as a promising treatment approach for removing EC from contaminated water by *in situ* generation of various reactive oxygen species (ROS).<sup>4</sup> For example, the Fenton process is the most widely used AOP process for removing EC from drinking water and wastewater. Among various Fenton processes, Fenton (Fe<sup>2+</sup>/H<sub>2</sub>O<sub>2</sub>), photo-Fenton (Fe<sup>2+</sup>/H<sub>2</sub>O<sub>2</sub>/light), and Fenton-like (Fe<sup>3+</sup>/H<sub>2</sub>O<sub>2</sub>) processes are environment friendly due to their low or nontoxicity, cheap, and easy operation.<sup>5,6</sup>

The generated ROS can quickly oxidize EC to biodegradable intermediates and finally completely being removed as mineralized products (e.g., CO<sub>2</sub> and H<sub>2</sub>O). However, the conventional homogeneous Fenton process is not cost-effective and sustainable due to the higher generation of iron sludge by consuming higher soluble iron and H<sub>2</sub>O<sub>2</sub>.<sup>7</sup>

The heterogeneous Fenton process may overcome the limitation of the conventional process by reducing iron sludge and maintaining its pH to neutral.<sup>8</sup> Iron-containing materials such as zerovalent iron (nZVI),<sup>9</sup> iron oxide,<sup>10,11</sup> and iron sulfide<sup>12</sup> are commonly tested as heterogeneous Fenton catalysts.<sup>7,13</sup> However, the synthesized form of these materials

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is susceptible to aggregate and form large particles in the presence of  $O_2$  and  $H_2O$ , which could result in loss dispersibility and the specific area, thus ultimately reducing their catalytic activity. Consequently, a particular precaution is required to keep the catalyst stable, uniform, and well-distributed in the reaction medium. Pyrite ( $FeS_2$ ) is one of the most plentiful natural iron mineral on Earth, and its synthesized form has proven its excellent catalytic activity<sup>12</sup> but its weak stability is also observed in the presence of air and water.<sup>14</sup> Light irradiation may also weaken its stability. To keep the pyrite chemically stable, a protecting layer on the surface of pyrite is a crucial step.

Graphitic carbon nitride ( $g-C_3N_4$ ), reduced graphene oxide (rGO), and graphene oxide (GO) are widely investigated promising materials to stabilize hybrid materials. Of these, graphene oxide has a high electric potential density, electron mobility, optical absorption, and high impermeability.<sup>14,15</sup> Moreover, its oxygen-containing functional groups and larger specific surface area with pyrite<sup>16</sup> are required for photocatalytic applications.<sup>17</sup> Although natural or synthetic pyrite has been used to degrade some of the chemical contaminants,<sup>8,18</sup> no studies have evaluated the feasibility of GO coated pyrite as a heterogeneous Fenton catalyst to simultaneously eliminate both biological and chemical contaminants.

This study aims to develop a new heterogeneous photo-Fenton catalyst for eliminating ARB, ARGs, and MPs. First, the nano  $FeS_2$  was synthesized *via* a facile and solvothermal process. The synthesized nanomaterials were coated with  $g-C_3N_4$ , rGO, and GO by an ultrasonication process. After that, we evaluated the ARB inactivation efficiency of these Fenton catalysts. Finally, we selected the  $FeS_2@GO$  photocatalyst for eliminating biological and chemical hazards. To this end, *Escherichia coli* containing  $\beta$ -lactam and tetracycline resistance genes was used as a biological model, while four pharmaceuticals, including sulfamethoxazole (SMX), carbamazepine (CBZ), diclofenac (DCF), and mecoprop (MCP), were chosen as chemical contaminants. To evaluate the fundamental reaction mechanisms of catalyst, both chemical scavengers and electron paramagnetic resonance (EPR) were used to confirm the contribution of reactive species to degrade these contaminants. Additionally, various water matrixes were used to validate the efficiency of heterogeneous catalyst as well as its stability and reusability. This research demonstrated that the  $FeS_2@GO$ -based heterogeneous system is a promising “one-stop” treatment process for the removal of EC including biological and chemical contaminants.

## MATERIALS AND METHODS

**Chemicals.** A complete list of chemicals is provided in Text S1 in the Supporting Information (SI).

**Synthesis of  $FeS_2$  with Different Coated Materials and Structure Characterization.** The  $FeS_2$  was obtained by a solvothermal method. Typically, 7.95 g of  $FeSO_4 \cdot 7H_2O$  (25 mM) and 3.95 g of  $Na_2S_2O_3$  (25 mM) were mixed in 30 mL of ethylene glycol. After that 400 mg of sulfur (S) (12.5 mM) was added to the solution and mixed properly. Finally, the suspension was placed in a Teflon-lined stainless-steel autoclave (100 mL) and reacted for 6 h at 200 °C. Nano pyrite was acquired as a black precipitate. The precipitate was washed by deionized water, diluted sulfuric acid ( $H_2SO_4$ ), and ethanol several times until the filtrate was colorless and the pH was neutral. Eventually, the obtained black precipitate was

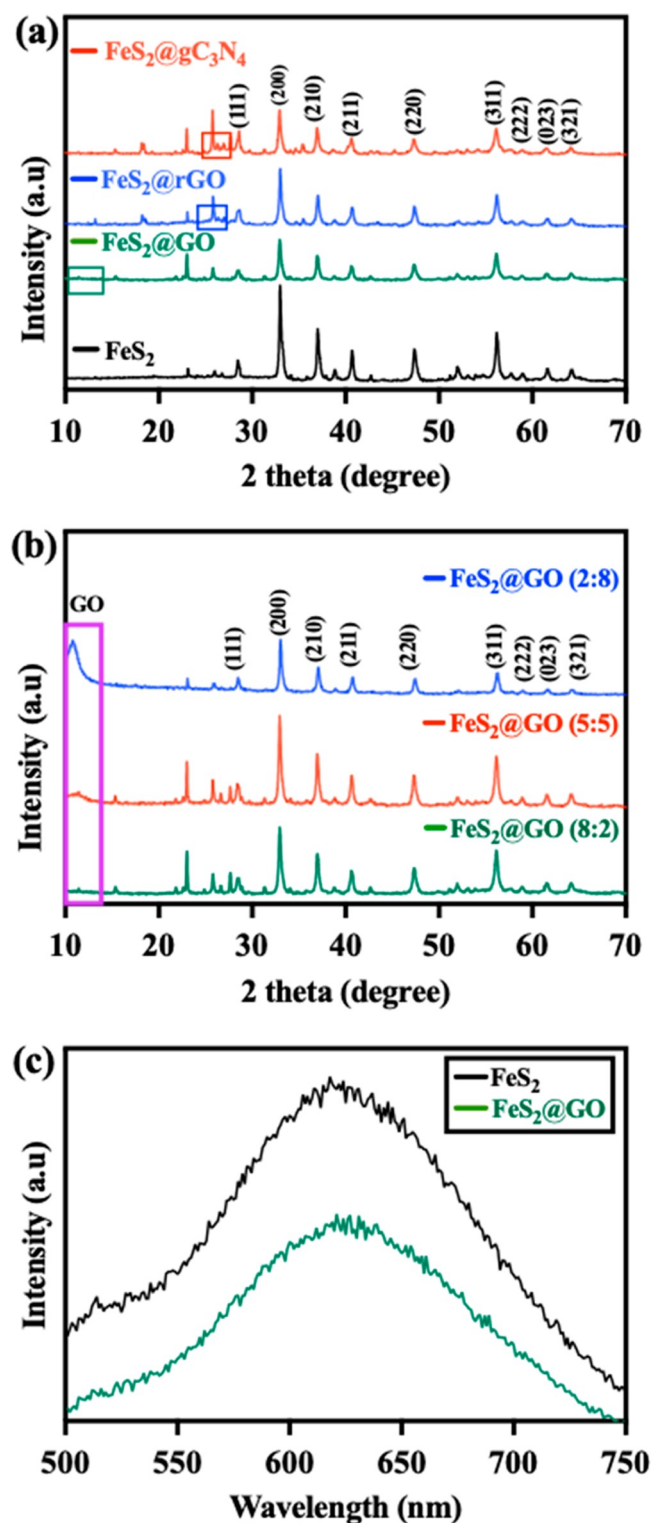
dried overnight at 55 °C. A series of composite materials were prepared by mixing with coated materials ( $g-C_3N_4$ , rGO, and GO) by an ultrasonication process. Typically, 40 mg of coated materials was dissolved in a 7:3 mL water and isopropanol solution and sonicated for 30 min. After that, an appropriate amount of nano  $FeS_2$  was mixed with the above solution to get the required ratio (8:2, 5:5, and 2:8) and sonicated again for 1 h. After sonication, the resultant composites were washed several times with distilled water and ethanol, and dried overnight at 55 °C. The structure and morphology of the synthesized nano  $FeS_2$  and their hybrid composites  $FeS_2@g-C_3N_4$ ,  $FeS_2@rGO$ , and  $FeS_2@GO$  catalysts were analyzed by various methods. The specific characterization methods are shown in Text S2.

**Catalytic Activity Measurement.** The heterogeneous photo-Fenton process was carried out in a 50 mL reactor for the elimination of emerging contaminants (ARB, ARGs, and MPs) under solar light. Before selecting the efficient catalyst, various inactivation experiments against ARB under different conditions ( $Fe_3O_4$ , nZVI,  $FeS_2$ ,  $g-C_3N_4$ , rGO, GO,  $FeS_2@g-C_3N_4$ ,  $FeS_2@rGO$ , and  $FeS_2@GO$ ) were separately examined. For each of the experiments, the required volume of selected material and  $H_2O_2$  were added in a 50 mL synthetic wastewater (SWW) matrix containing  $\sim 10^6$  CFU/mL antibiotic resistant *E. coli* under continuous stirring (250 rpm). The initial pH for all the reactions was neutral pH ( $7.40 \pm 0.05$ ). Prior to irradiation, the required amount of catalyst was homogeneously dispersed in the reactor for 10 min to achieve an adsorption–desorption equilibrium. Subsequently, a required volume of  $H_2O_2$  was added to the reaction medium and switched on the light under continuous stirring. The reactor was placed under simulated sunlight irradiation generated by PLS-SXE300 (Beijing Perfectlight Technology Co. Ltd., Beijing, China), using a 138-xenon lamp (200–800 nm, 300W), with an average light intensity of 130 mW/cm<sup>2</sup>. The distance between the lamp and the surface of the reaction medium was 12 cm. The required volumes of the sample were collected for ARB, e-ARG, intracellular ARG (i-ARG), and MP after certain time interval. For ARB calculation, 100  $\mu$ L of treated samples was diluted with SWW matrix and poured onto a selective LB agar plate. After 24–36 h of incubation at 37 °C, the number of viable cells was counted according to the previous study.<sup>19</sup> The  $FeS_2@GO$  composite was separated from the reaction medium by centrifugation. The details of ARB enumeration and bacterial regrowth were explained in our previous studies<sup>19–21</sup> as well as Texts S3 and S4. For ARG, the extraction of DNA and quantification of respective genes by real-time PCR (RT-PCR) analysis were described in Texts S5 and S6 according to our previous studies.<sup>19,20</sup> The standard curve of each gene is illustrated in Figure S1. The length of the primer of each targeted gene is shown in Table S1.

**Analytical Methods.** The detailed analytical methods for standard preparation and quantification of MPs are described in our previous studies<sup>20,22</sup> as well as Text S7.

## RESULTS AND DISCUSSION

**Structural and Compositional Characterizations.** The structure and morphology of the synthesized nano  $FeS_2$  and their hybrid composites  $FeS_2@g-C_3N_4$ ,  $FeS_2@rGO$ , and  $FeS_2@GO$  catalysts were characterized using various methods. The crystal phase and structural information was investigated by X-ray diffraction (XRD) patterns (Figure 1). The synthesized pyrite has nine sharp base peaks at 28.70°,



**Figure 1.** XRD spectra of (a) pyrite and its coated composites. (b) Different ratios of FeS<sub>2</sub>/GO. (c) PL spectra of FeS<sub>2</sub> and FeS<sub>2</sub>@GO (8:2) with an excitation wavelength of 450 nm.

33.02°, 37.07°, 40.81°, 47.49°, 56.30°, 60.24°, 62.83°, and 65.44° corresponding to the (1 1 1), (2 0 0), (2 1 0), (2 1 1), (2 2 0), (3 1 1), (2 2 2), (0 2 3), and (3 2 1) faces, respectively.<sup>14,23</sup> The sharp diffraction XRD peaks suggest their high hump range from 10° to 25° (2-θ). A broad and weak peak in the XRD pattern is observed at 11.22°, 25.90°, and 25.82° for GO, rGO, and g-C<sub>3</sub>N<sub>4</sub>, respectively. The intensity of

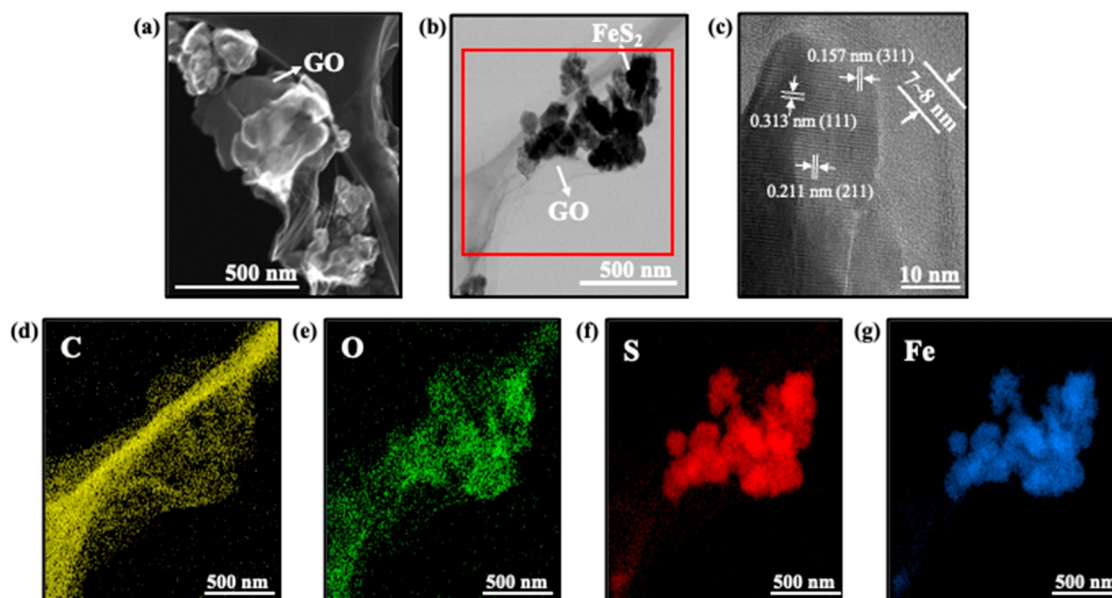
the coated materials is faded as compared to that of FeS<sub>2</sub>, suggesting the coated layer was formed on the surface of FeS<sub>2</sub> (Figure 1a). However, the diffraction pattern of GO is increased at 11.22° with the increase of its concentration in the hybrid material, and a high-intensity peak is observed at the ratio of 2:8 (Figure 1b).

In order to examine the generation, separation, and recombination of charge carriers, the photoluminescence (PL) spectra of FeS<sub>2</sub> and FeS<sub>2</sub>@GO were measured (Figure 1c). The FeS<sub>2</sub>@GO displays much lower PL intensities than FeS<sub>2</sub>. The result confirms that the GO coated surface is hindered the recombination of  $e^-$  and  $h^+$  effectively. This result also confirms the transmission of a photoexcited electron from the surface of FeS<sub>2</sub> catalyst to GO layer and confines the recombination of electrons and holes.<sup>14,24</sup>

In order to further analyze the morphology of the synthesized FeS<sub>2</sub> and FeS<sub>2</sub>@GO, scanning transmission electron microscopy (STEM) was conducted (Figure 2a). The nano pyrite is well-covered by a thick envelope with the coated material (GO). The resulting nanoparticles are characterized with a size of 100–200 nm. The TEM images further verify more directly that the FeS<sub>2</sub> nanoparticles are coated perfectly in the GO monolayers sheet (Figure 2b). GO sheets form confined spaces with large internal voids on the surface of FeS<sub>2</sub> particles. The high-resolution transmission electron microscopy (HRTEM) image of FeS<sub>2</sub>@GO is presented in Figure 2c. The interplanar distances of 0.313, 0.221, and 0.157 nm correspond to the (111), (211), and (311) lattice planes of FeS<sub>2</sub>@GO, respectively, which are consistent with the XRD results. However, 7–8 nm thick wrinkles were observed on the FeS<sub>2</sub>@GO. It might be the coating layer of GO. The energy-dispersive X-ray spectroscopy (EDS) confirms the existence of C, O, Fe, and S elements in the red square marked region of the TEM image as shown in Figure 2b. The EDS mapping also confirms that the synthesis of FeS<sub>2</sub>@GO was done by simply mixing of C, O, S, and Fe elements (Figure 2d–g) while pyrite only contained the mixer of S and Fe elements (Figure S2d–g). This implies that there is no coated GO layer.

The X-ray photoelectron spectroscopy (XPS) spectra were also examined to confirm the chemical state of the compositional elements of FeS<sub>2</sub>@GO (Figure 3). The global profile (Figure 3a) of FeS<sub>2</sub>@GO shows clear characteristic peaks corresponding to S 2p, C 1s, O 1s, and Fe 2p, which are located at 163.0, 284.8, 531.0, and 707.17 eV, respectively.<sup>14</sup> The carbonyl (C=O) and carboxyl (O=C=O) groups peaks appear at 286.4 and 288.1 eV, respectively (Figure 3b).<sup>14</sup> The hydroxyl (H–O) and water (H<sub>2</sub>O) peaks appear at 531.0 and 533.5 eV, respectively, which correspond to the oxygen functional groups (Figure 3c).<sup>14</sup> The bimodal features of the S 2p spectrum (S 2p<sub>3/2</sub> and S 2p<sub>1/2</sub>) were found at 162.44 and 163.57 eV, respectively (Figure 3d), with a splitting of 1.13 eV, and are attributed to disulfide (S<sub>2</sub><sup>2-</sup>) and monosulfide (S<sup>2-</sup>) dimer.<sup>25,26</sup> The peaks around 707.1 and 720.0 eV in the Fe 2p spectrum are assigned to Fe 2p<sub>3/2</sub> and 2p<sub>1/2</sub>, respectively, and correspond to Fe<sup>2+</sup>-S (Figure 3e). The peak around 709.5 eV corresponds to Fe<sup>2+</sup>-O.<sup>14,27</sup> A new peak around 711.8 eV is also observed with the previous three peaks in GO-coated FeS<sub>2</sub> (Figure 3e). This is attributed to the formation of Fe<sup>3+</sup>-O due to the oxidation of Fe<sup>2+</sup>-O on the surface of the GO–pyrite.<sup>14</sup> The carbonyl bond (C=O) of GO and the Fe<sup>3+</sup>-O of FeS<sub>2</sub> may form covalent bonds with each other, which can help to form a stable GO coated layer on the surface of pyrite.





**Figure 2.** Characterization of  $\text{FeS}_2$ @GO (8:2) composites: (a) STEM, (b) TEM, (c) HRTEM images of  $\text{FeS}_2$ @GO, and (d–g) EDS elemental mappings images of  $\text{FeS}_2$ @GO.

Collectively, these characterization analyses confirmed that nano  $\text{FeS}_2$  with GO cover was successfully synthesized, and the uniform coating of GO may act as a protective film of pyrite nanomaterials. In contrast, the XPS spectra of  $\text{FeS}_2$  showed three major peaks for the Fe 2p spectrum in Figure 3f.

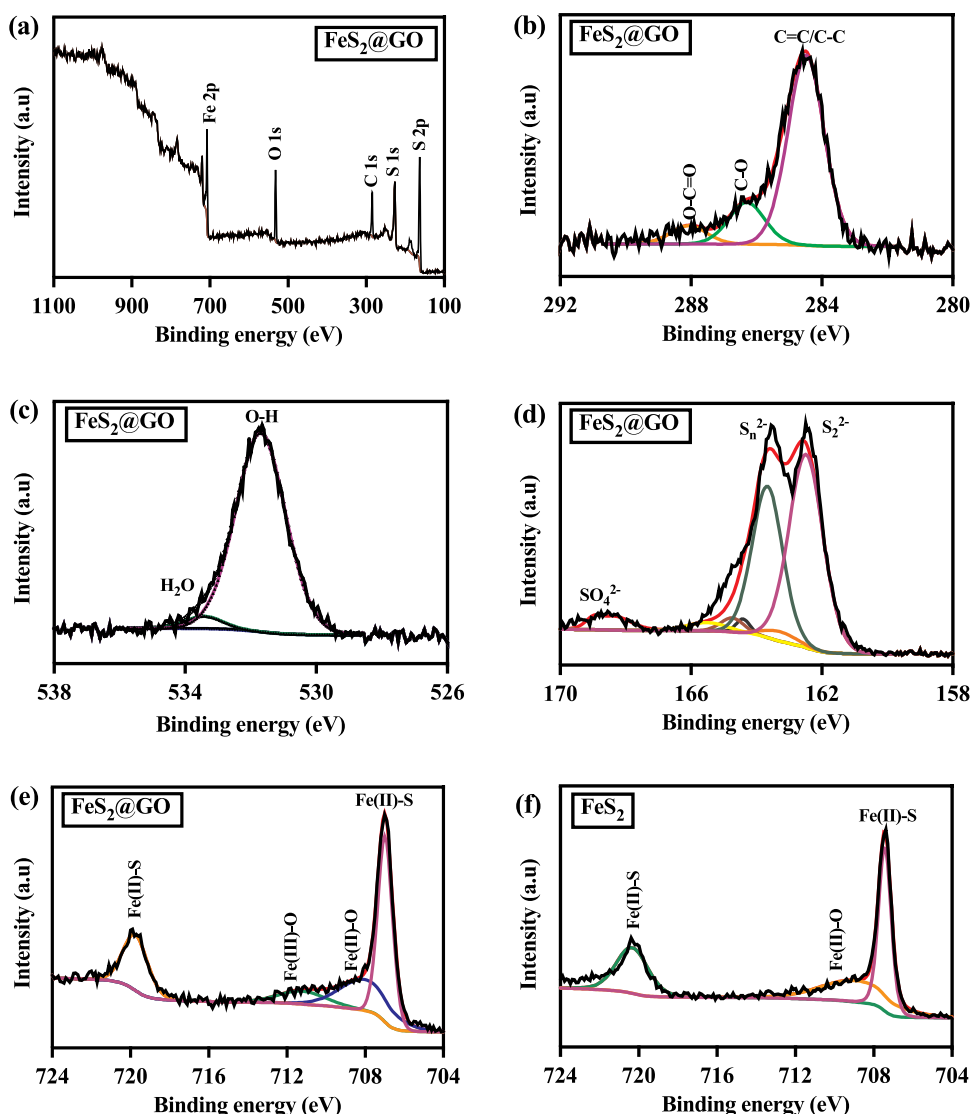
**ARB Inactivation and Regrowth Potential.** In order to assess ARB inactivation efficiency of various heterogeneous catalysts, the abundance of *E. coli* was enumerated in the SWW matrix. We first synthesized three common nanomaterials ( $\text{Fe}_3\text{O}_4$ , nZVI, and  $\text{FeS}_2$ ) and compared their inactivation performance. As shown in Figure S4a, the bare  $\text{FeS}_2$  was more efficient, as compared to other two catalysts, and evidenced a complete ARB inactivation under 30 min of light irradiation. The excellent disinfection performance attained by  $\text{FeS}_2$  might be due to enhanced circulation between  $\text{Fe}^{3+}$  and  $\text{Fe}^{2+}$  ions. Consequently,  $\text{FeS}_2$  was selected as a primary compound and coated with g- $\text{C}_3\text{N}_4$ , rGO, and GO on its surface. Figure S4b illustrates the ARB inactivation performance of the selected composites by using 0.25 g/L of each composite (8:2) and 1.0 mM  $\text{H}_2\text{O}_2$ .  $\text{FeS}_2$ @g- $\text{C}_3\text{N}_4$  and  $\text{FeS}_2$ @rGO were only able to achieve 3.88-log and 4.97-log reduction, respectively. In contrast, the  $\text{FeS}_2$ @GO catalyst was able to reach the detection limit after 30 min of treatment.

The control experiments (Figure 4a) were examined to further compare impacts of bare material and/or synthesized composites on ARB inactivation, and the data is shown in Table S2. In the dark Fenton process ( $\text{FeS}_2$ @GO and  $\text{H}_2\text{O}_2$ ), the inactivation of ARB was very low (3.60-log). On the contrary, only  $\text{H}_2\text{O}_2$  increased the ARB inactivation efficiency in the presence of light (4.83-log) but it was unable to reach the detection limit after 30 min of treatment. In this heterogeneous process, only catalyst or  $\text{H}_2\text{O}_2$  was not sufficient, but the synergistic effect of  $\text{FeS}_2$ @GO with  $\text{H}_2\text{O}_2$  under light can eliminate all ARB.

Finally,  $\text{FeS}_2$ @GO catalyst was selected as a heterogeneous photo-Fenton catalyst for the removal of biological and chemical hazards. Three molar ratios of  $\text{FeS}_2$ @GO (8:2, 5:5, and 2:8), three doses (0.12, 0.25, and 0.38 g/L) of  $\text{FeS}_2$ @GO, and three concentrations of  $\text{H}_2\text{O}_2$  (0.5, 1.0, and 1.5 mM) were

primarily examined to get the optimal reaction conditions of the  $\text{FeS}_2$ @GO catalyst (Figure 4b–d). The molar ratio of  $\text{FeS}_2$ /GO determines the stability and photocatalytic activity, thus affecting the bacterial inactivation efficiency. With decreasing the molar ratio of  $\text{FeS}_2$ /GO, the material tended to be more stable, thus resulting in lower disinfection efficiency. Specifically, the efficiency was reduced from 6.35- to 4.14-log for the molar ratio of 8:2 to 5:5, respectively (Figure 4b), while the inactivation efficiency was dropped to 3.71-log at the ratio of 2:8 in  $\text{FeS}_2$ @GO and the particles appeared as clearly defined and discrete suspensions in the reaction medium. The solution pH was also dropped from  $\sim 7.4 \pm 0.2$  to  $6.5 \pm 0.2$  at the molar ratio of  $\text{FeS}_2$ @GO (2:8). This might be due to the presence of various carboxyl ( $-\text{COOH}$ ) and hydroxyl groups ( $-\text{OH}$ ) of GO, which can drop solution pH from neutral to acidic.<sup>28</sup> The highest removal (6.35-log) efficiency was received at an 8:2 ratio after 30 min of treatment.

$\text{H}_2\text{O}_2$  shows a key role in this process because it significantly affected the inactivation efficiency. As shown in Figure 4c, we observed around 4.0-log ARB reduction after 30 min of reaction at a concentration of 0.5 mM  $\text{H}_2\text{O}_2$ . With increasing  $\text{H}_2\text{O}_2$  concentration, the inactivation efficiency increased and reached the detection limit of ARB at a higher concentration (1.0 mM) with 30 min of contacting time. In contrast, a shorter treatment time (20 min) was required for 1.5 mM  $\text{H}_2\text{O}_2$ . The  $\text{H}_2\text{O}_2$  molecule is much smaller (1.2–1.4 Å) than the bacterial porins (10–11 Å).<sup>29</sup> Due to the similar polarity of  $\text{H}_2\text{O}_2$  to water, it could diffuse into the cell, potentially initiating cell death. As a strong oxidizing agent, it may oxidize the cysteinyl and methionine protein and cell lipids, thus finally deactivating the bacterial cell.<sup>30</sup> Moreover, the presence of higher  $\text{H}_2\text{O}_2$  (1.5 mM) deteriorated the solution pH since the dissolution rate  $\text{H}_2\text{O}_2$  in the presence of pyrite is much faster than the molecular oxygen in solution,<sup>28</sup> which could lead to a higher ARB inactivation efficiency under weakly acidic conditions (pH of 4.85). Therefore, the 1.0 mM  $\text{H}_2\text{O}_2$  concentration was selected as the optimal ratio for eliminating the hazards.



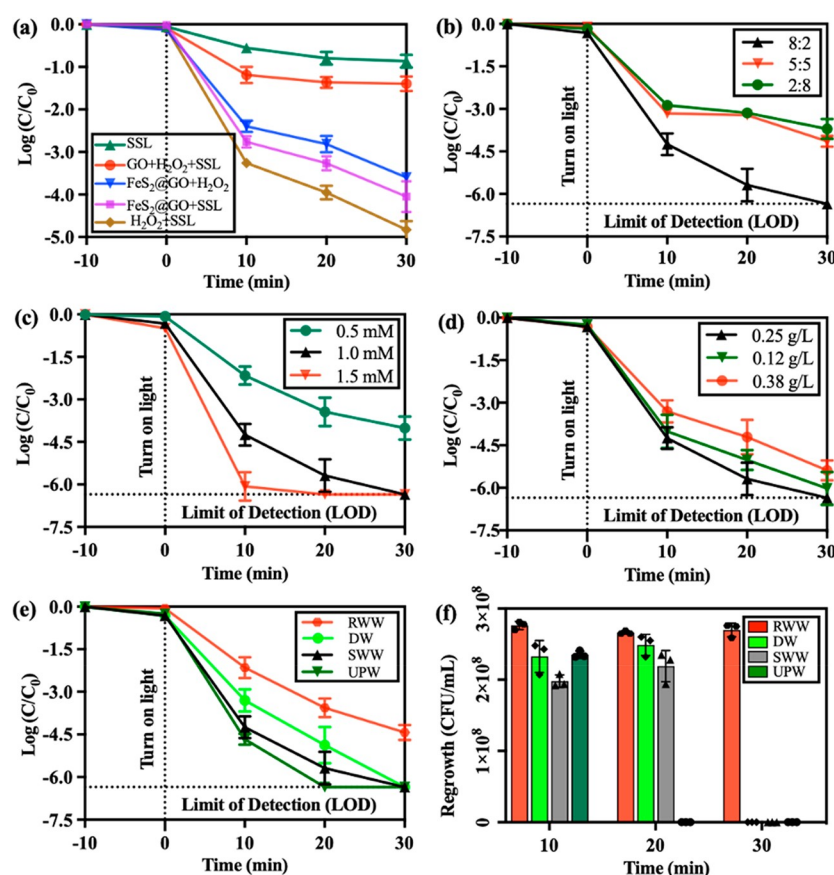
**Figure 3.** XPS spectra of  $\text{FeS}_2@\text{GO}$ : (a) global scan, (b) C 1s, (c) O 1s, (d) S 2p, (e) Fe 2p, and (f) Fe 2p for  $\text{FeS}_2$ .

The inactivation efficiency was also examined by changing the catalyst dosage from 0.12 to 0.38 g/L. As illustrated in Figure 4d, the ARB inactivation efficiency increased from 6.02-log to 6.35-log with the increase of catalyst doses from 0.12 to 0.25 g/L during 30 min of treatment. The higher disinfection efficiency is due to the higher active sites of catalyst that may generate more ROS, which can help to kill bacteria.<sup>31</sup> However, a higher catalyst dose (0.38 g/L) reduced the inactivation efficiency (5.39-log). As depicted in Figure 4d, 0.25 g/L  $\text{FeS}_2@\text{GO}$  catalyst was efficient for killing all the initial bacterial cells ( $>10^6$  CFU/mL). Taking both economy and efficiency into consideration, the optimal doses consisting of 0.25 g/L catalyst, 1 mM  $\text{H}_2\text{O}_2$  (34.2 mg/L), and 8:2 ratio of  $\text{FeS}_2@\text{GO}$  were employed as the optimal condition throughout all the experiments below.

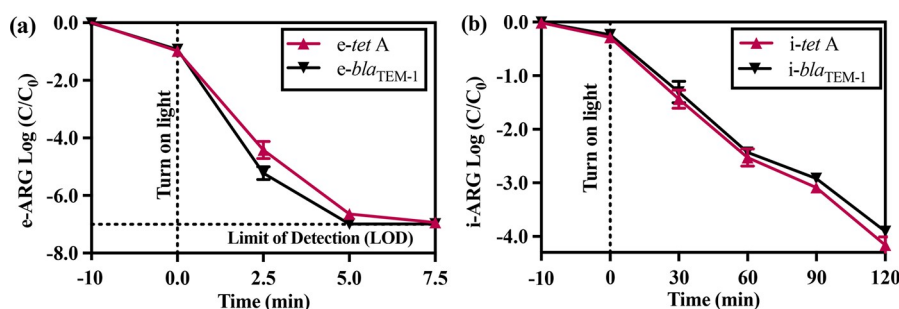
This study further examined ARB inactivation under different water matrixes, including ultrapure water (UPW), drinking water (DW), synthetic wastewater (SWW), and real wastewater effluent (RWW). At the optimal doses, the heterogeneous photo-Fenton process reached the detection limit (6.35-log reduction) after 30 min treatment in three water matrixes, including UPW, DW, and SWW. In contrast, a

lower inactivation activity (4.43-log reduction) was observed in RWW (Figure 4e). This might be associated with dissolved organic matter (DOM) in RWW. DOM could result in sensitizing and quenching the photooxidation process.<sup>32</sup> In addition, bicarbonates also act as ROS scavengers,<sup>33</sup> thus decreasing disinfection efficiency.

The regrowth potential of ARB at different water matrixes was also evaluated in order to establish the connection between the treatment time and bacterial cell damage, thus avoiding the regrowth of the bacteria after the treatment (Figure 4f). With the increase of treatment time, the regrowth activity was decreased and indicated that the treatment time played a significant role in killing the bacteria from three water matrixes (UPW, DW, and SWW). After 30 min of treatment, no ARB grew (0 CFU/mL) in the above three water matrixes. However, bacterial regrowth happened in the SWW and DW matrixes when treating contaminated water for 20 min after 24 h of incubation (Figure S4c). The regrowth became more obvious after 48 h of incubation (Figure 4f). In addition, due to incomplete ARB inactivation, the remaining bacteria (around 2.0-log) were alive in the matrix, which can further regrow during the 24 or 48 h incubation periods.



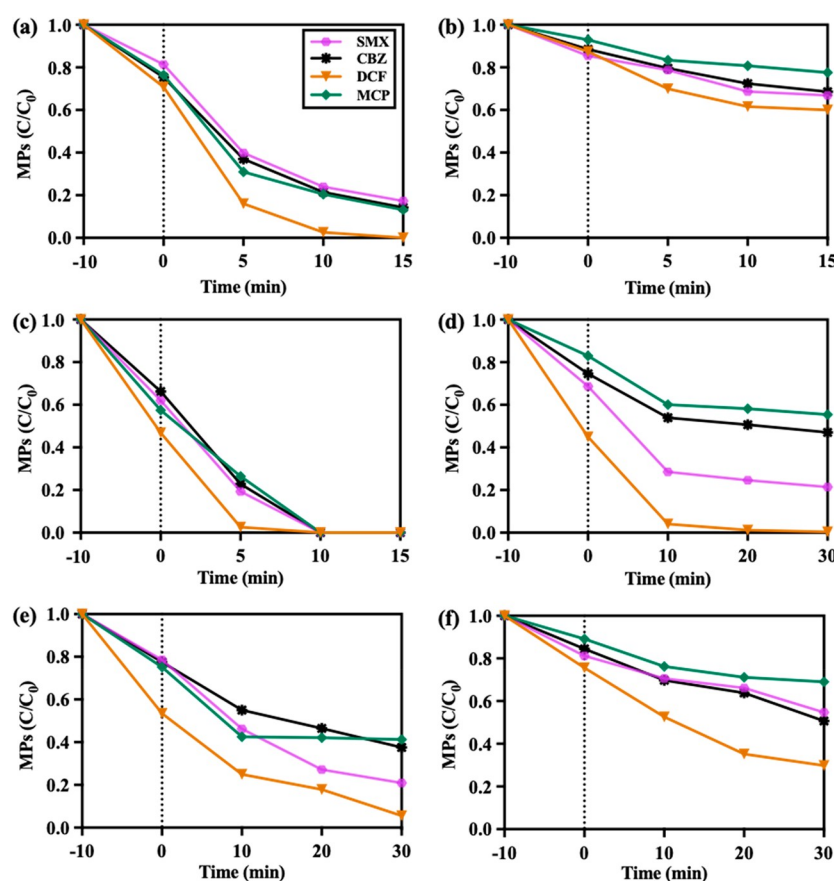
**Figure 4.** Inactivation of ARB in synthetic wastewater matrix (a) performance of respective controls, the performance of (b) different  $\text{FeS}_2/\text{GO}$  ratios, (c)  $\text{H}_2\text{O}_2$  concentrations, (d) catalyst dosages, (e) inactivation of ARB in different water matrixes, and (f) regrowth activity at 48 h of incubation under the heterogeneous photo-Fenton process. All experiments were conducted at neutral pH in the presence of  $10^6$  CFU/mL ARB, 0.25 g/L catalyst, 1 mM  $\text{H}_2\text{O}_2$  (34.2 mg/L), 8:2 ratio of  $\text{FeS}_2$  with coated materials, and  $130 \text{ mW}/\text{cm}^2$  light intensity. UPW, ultrapure water; DW, drinking water; SWW, synthetic wastewater; and RW, real wastewater effluent.



**Figure 5.** Degradation of ARGs (a) extracellular (e-ARGs) and (b) intracellular (i-ARGs) under heterogeneous photo-Fenton process in SWW matrix. All experiments were conducted at neutral pH in the presence of  $10^6$  CFU/mL ARB, 0.25 g/L  $\text{FeS}_2@\text{GO}$  (8:2), 1 mM  $\text{H}_2\text{O}_2$  (34.2 mg/L), and  $130 \text{ mW}/\text{cm}^2$  of light intensity.

**Degradation of e-ARG and i-ARG.** During the disinfection process, ARB loses membrane integrity, thereby releasing cytoplasmic contents and causing DNA to leak out of the cell.<sup>34</sup> These free DNA might still contain ARG, which could become sources to spread antibiotic resistance through horizontal gene transfer in bacteria in the environment.<sup>35</sup> Previous AOPs primarily focus on ARB inactivation, but ARG elimination has been overlooked.<sup>36–38</sup> In this study, RT-PCR analysis was carried out to determine the degradation of both e-ARG and i-ARG. The removal efficiency of e-ARGs (*tetA* and *bla*<sub>TEM-1</sub>) exhibited a similar removal trend over time (Figure 5a). Specifically, *bla*<sub>TEM-1</sub> promptly reached the

detection limit (7.0-log reduction) after 5 min of treatment as compared to 7.5 min for *tetA* degradation in the SWW matrix. The degradation rate of *bla*<sub>TEM-1</sub> was higher than that of *tetA* because it has  $\sim 1.5$  times higher T–T bases, which are more difficult to damage.<sup>19</sup> It is worth noting that the current study obtained a higher e-ARGs removal efficiency than that reported in previous studies.<sup>19,20</sup> For example, the homogeneous photo-Fenton process degraded  $\sim 5$ -log e-ARGs in the dose of  $\text{Fe}^{2+}$  and  $\text{H}_2\text{O}_2$  (0.178 and 0.55 mM).<sup>19</sup> In contrast, the EDDS modified photo-Fenton process degraded 6.0-log of e-ARGs (*tetA* and *bla*<sub>TEM-1</sub>) by 0.1:0.2:0.3 mM  $\text{Fe(III)}/\text{EDDS}/\text{H}_2\text{O}_2$  after 10 min of treatment in the SWW matrix.<sup>20</sup> The i-



**Figure 6.** Degradation profiles of four MPs in (a)  $\text{FeS}_2\text{@GO}$  under light in UPW, (b)  $\text{H}_2\text{O}_2$  under light in UPW, and degradation of MPs under heterogeneous photo-Fenton process in (c) UPW/ultrapure water, (d) SWW/synthetic wastewater, (e) DW/drinking water, and (f) RWW/real wastewater. All experiments were conducted using  $10\ \mu\text{g/L}$  of each MP,  $0.25\ \text{g/L}$   $\text{FeS}_2\text{@GO}$  (8:2),  $1\ \text{mM}$   $\text{H}_2\text{O}_2$  ( $34.2\ \text{mg/L}$ ), and  $130\ \text{mW/cm}^2$  of light intensity. SMX, sulfamethoxazole; CBZ, carbamazepine; DCF, diclofenac; MCP, mecoprop.

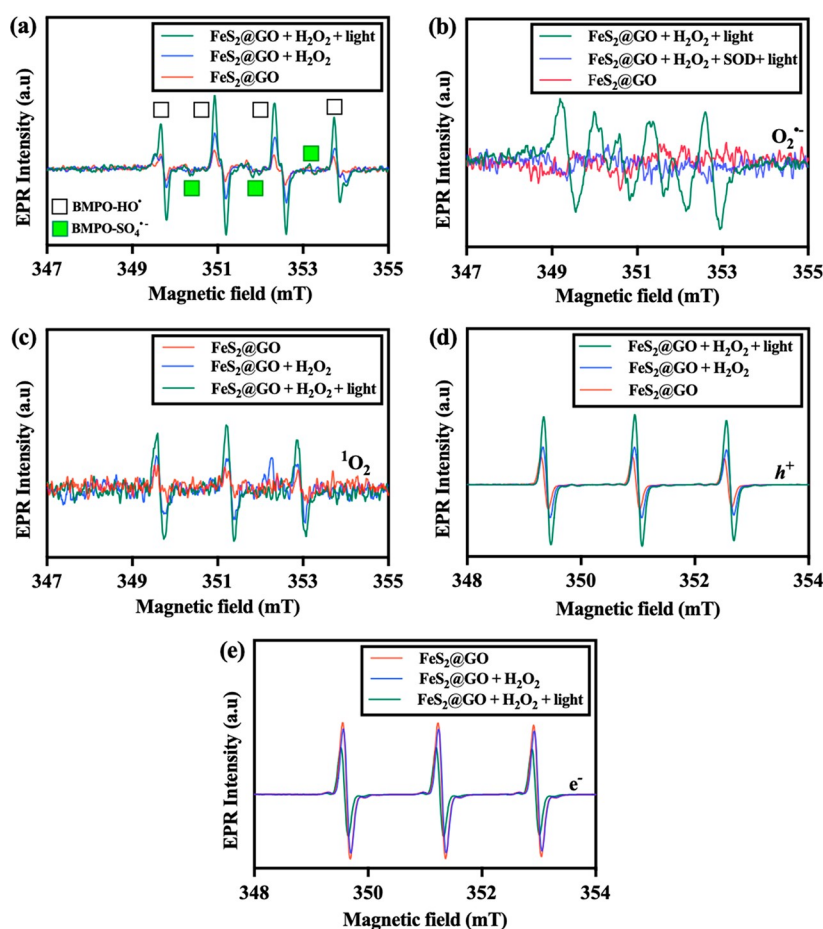
ARG removal efficiency was also explored (Figure 5b). Around 1.30-log of i-ARGs degradation was achieved after 30 min of treatment. With increasing the contacting time, the higher i-ARGs removal efficiency was detected, evidenced by 4.16-log of *tetA* and 3.91-log of *bla*<sub>TEM-1</sub> after 2 h of treatment, which are also higher compared to previous studies.<sup>19,20</sup>

**Degradation of Micropollutants.** To evaluate whether the current heterogeneous photo-Fenton process can efficiently degrade chemical hazards, the degradation efficiency of four typical MPs was quantified by LC-ESI-MS/MS. Only 80%–99% (Figure 6a) and 20%–40% (Figure 6b) of MPs were degraded in UPW water matrix due to the effect of  $\text{FeS}_2\text{@GO}$  or  $\text{H}_2\text{O}_2$  in the presence of light, respectively, during 15 min of treatment. In contrast, around 40%–50% of MPs were degraded in the dark condition (10 min) with the presence of  $\text{FeS}_2\text{@GO}$  catalyst (data not shown). Under the default dose ( $0.25\ \text{g/L}$   $\text{FeS}_2\text{@GO}$  and  $1\ \text{mM}$   $\text{H}_2\text{O}_2$ ), four chemical hazards were completely removed from the UPW matrix within 10 min of treatment (Figure 6c). A lower removal efficiency was recognized in DW, SWW, and RWW matrixes as compared to that of UPW. For example, approximately 99.6% DCF and 79.6% SMX were degraded, while CBZ and MCP showed lower than 45% degradation efficiency during 30 min of treatment in SWW (Figure 6d). This result suggests that the catalyst can degrade MPs but it cannot completely remove them from SWW and the similar trend also observed in the DW (Figure 6e) and RWW matrixes (Figure 6f). In the RWW matrix, the elimination efficiencies

were 31.0% MCP, 45.3% SMX, 49.3% CBZ, and 71.2% DCF during 30 min of treatment. These data indicate that the generation of reactive species was insufficient to completely degrade MPs, due to the presence of bicarbonates and DOM, which can scavenge ROS.<sup>33,39</sup> Thus, it requires to further optimize the dosage or contacting time when applying the heterogeneous photo-Fenton process to treat real wastewater effluent.

**Identification of Dominant Reactive Species.** The reactive species  $\text{HO}^\bullet$ ,  $\text{O}_2^{\bullet-}$ ,  $^1\text{O}_2$ ,  $h^+$ , and  $e^-$  were identified by using five chemical scavengers including isopropanol (IPA), superoxide dismutase (SOD), sodium azide (SA), triethanolamine (TEA), and potassium dichromate ( $\text{K}_2\text{Cr}_2\text{O}_7$ ), respectively. The detailed identification procedure of reactive species is described in Text S8. In the absence of scavengers, four of examined chemical hazards were completely removed from the UPW matrix within 10 min of treatment (Figure 6c). When TEA ( $h^+$  scavenger) was added to the reaction medium, 47%–52% of MPs removal was observed (Figure S5a). The effects of TEA on MPs removal collectively indicate that  $h^+$  might show a key contribution in the heterogeneous process. Isopropanol was used to scavenge  $\text{HO}^\bullet$  radicals, resulting in an obvious decrease in MPs removal efficiency. As illustrated in Figure S5b, around 20% inhibition was observed for MPs degradation after 15 min of treatment in UPW. The scavenger sodium azide was used to scavenge singlet oxygen ( $^1\text{O}_2$ ). Compared to  $\text{HO}^\bullet$  radicals,  $^1\text{O}_2$  showed a minor effect on removing MPs. The highest inhibition effect of SA was





**Figure 7.** EPR spectra of FeS<sub>2</sub>@GO: (a) detection of HO• in the presence of BMPO (0.5 mM), (b) O<sub>2</sub>•<sup>-</sup> in the presence of BMPO (0.5 mM), (d) <sup>1</sup>O<sub>2</sub> in the presence of TEMP (4.0 mM), (d) h<sup>+</sup> in the presence of CPH (0.1 mM), and (e) e<sup>-</sup> in the presence of TEMPO (0.1 mM). The initial catalyst dose was 0.25 g/L FeS<sub>2</sub>@GO (8:2), 1 mM H<sub>2</sub>O<sub>2</sub>, and 130 mW/cm<sup>2</sup> of light intensity during experiments.

observed for MCP and SMX, and it inhibited around 56.3% and 39.0%, respectively, while the lowest inhibition was seen for DCF and CBZ (Figure S5c). Although the redox potentials of <sup>1</sup>O<sub>2</sub> and O<sub>2</sub>•<sup>-</sup> were lower than that of the HO• radical, the latter <sup>1</sup>O<sub>2</sub> can also degrade contaminants.<sup>40</sup> Using superoxide dismutase (SOD) to quench O<sub>2</sub>•<sup>-</sup>, the result showed that O<sub>2</sub>•<sup>-</sup> contributed a minor role to MPs degradation (Figure S5e). As an electron scavenger, K<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub> was used to determine its role on MPs degradation, and it showed a minor inhibition effect in four of examined MPs degradation (Figure S5d). To better understand the roles of methanol and its impact during the heterogeneous photo-Fenton process, we used it to scavenge simultaneously h<sup>+</sup> and HO•. The highest inhibition effect of methanol was observed for MPs and it was around 25%–60.7%, and DCF inhibition was insignificant (Figure S5f). In this present study, IPA and TEA were also used to scavenge HO• and h<sup>+</sup>, respectively. The chemical scavenger TEA also confirms its dominant inhibition characteristic for MPs degradation, which indicates that the hole shows a dominant role in this treatment process. The hole can degrade the hazards in two ways: (1) directly by reacting with organic matters, and (2) indirectly by generating various ROS. However, direct degradation was not significant with or without light. The indirect generation of ROS in the presence of a hole may show a vital role in degrading both chemical and biological hazards. The dominant radical HO•, which can also

be generated by the hole, is highly oxidative and responsible for MPs degradation.<sup>41</sup>

To further identify the generation of these reactive species in this heterogeneous process, the EPR spectra were recorded to confirm the contributor of various reactive species by adding trapping agents like CPH, TEMPO, TEMP, and BMPO before and during solar irradiation (Figure 7), and the detail procedure is explained in Text S9. In the presence of FeS<sub>2</sub>@GO catalyst and trapping agent, characteristic EPR signal showed a very lower intense signal for <sup>1</sup>O<sub>2</sub> and O<sub>2</sub>•<sup>-</sup> under dark conditions (red line spectrum), while a low-intensity four-line spectrum with the intensity ratio of 1:2:2:1 was observed, which corresponds to the BMPO/HO• (Figure 7a, red line).<sup>42</sup> This suggests that HO• radicals can degrade MPs in the absence of light, evidenced by 33%–55% MPs degradation in absence of light (dark condition). After adding H<sub>2</sub>O<sub>2</sub> in the reaction medium, a high-intensity EPR signal was observed due to the reaction of H<sub>2</sub>O<sub>2</sub> with the FeS<sub>2</sub>@GO catalyst (i.e., Fenton reaction, blue line spectrum), but the intensity was more obvious under light irradiation (green line spectrum). To confirm the generation of O<sub>2</sub>•<sup>-</sup> in a heterogeneous process, methanol was used as the reaction medium instead of water. As shown in Figure 7b, the EPR spectra indicate BMPO/O<sub>2</sub>•<sup>-</sup> spin adduct, thus validating the generation of O<sub>2</sub>•<sup>-</sup>. We added the SOD after 3 min of irradiation and observed low-intensity EPR spectra (Figure 7b, blue line), which correspond to O<sub>2</sub>•<sup>-</sup> radicals.<sup>43</sup> In the presence of TEMP, we noticed a three-line



spectrum with an intensity ratio of 1:1:1, confirming the generation of  $^1\text{O}_2$  (Figure 7c). In the presence of light, a bit higher intense signal was observed, confirming the generation of  $^1\text{O}_2$  in the  $\text{FeS}_2\text{@GO}$ -based Fenton process. We also observed a three-line spectrum with the same intensity ratio of 1:1:1 in the presence of the CPH probe, confirming the generation of  $h^+$  (Figure 7d). The spectrum was more intense and remarkable under solar irradiation (green line spectrum), resulting from the higher generation of  $h^+$ . A similar three-line spectrum was observed for trapping electrons with TEMPO in Figure 7e. High-intensity spectra confirm the lower amount of trapping electrons. Under solar irradiation, more iron ions are reduced and comparatively higher electrons were free from its surface. Thus, the characteristic EPR spectra were more flattened (green line) in the  $\text{FeS}_2\text{@GO}$ -based photo-Fenton process. Collectively, on the basis of the chemical scavenger test and EPR measurement, we conclude that the  $\text{HO}^\bullet$  radical plays a significant role in eliminating contaminants, followed by  $^1\text{O}_2$  and  $\text{O}_2^{\bullet-}$ .

**Proposed Reaction Mechanisms.** On the basis of the above results, underlying mechanisms for ARB inactivation and MPs degradation were proposed. In general, the contaminants removal efficiency can be attributed to several ROS, including  $\text{HO}^\bullet$ ,  $^1\text{O}_2$ , and  $\text{O}_2^{\bullet-}$  radicals. In the presence of simulated solar light, the photoexcited  $e^-$  can transmit from the surface of  $\text{FeS}_2$  catalyst to GO layer and confines the recombination of  $e^-$  and  $h^+$ . The photoinduced  $e^-$  reduced the  $\text{Fe}^{3+}$  ion from the complex to give  $\text{Fe}^{2+}$ . After that, a set of cyclic reactions occur, which can generate various ROS. The oxidative cycle reaction initiates with the decomposition of  $\text{H}_2\text{O}_2$  through the  $\text{Fe}^{2+}$  and followed by the generation of  $\text{O}_2^{\bullet-}$ .  $\text{O}_2^{\bullet-}$  can also enhance the reduction of  $\text{Fe}^{3+}$  to  $\text{Fe}^{2+}$  and can oxidize into  $\text{HO}^\bullet$  and  $\text{HO}^-$ . The  $\text{HO}^\bullet$  radical is the dominant ROS in this process, which can mainly degrade the contaminants. However, its disproportionation reaction can generate  $^1\text{O}_2$ . Therefore, the chain reaction continues until a complete degradation and the light can enhance the degradation process by generating more ROS such as  $\text{HO}^\bullet$ ,  $^1\text{O}_2$ ,  $\text{O}_2^{\bullet-}$ , and  $\text{H}_2\text{O}_2$ .

**Catalytic Stability of  $\text{FeS}_2\text{@GO}$ .** The  $\text{FeS}_2\text{@GO}$  photo-Fenton process has no noticeable degradation after five consecutive cycles (Figure 8). In the first two cycles, all the hazards reached the detection limit after 10 and 15 min of treatment, respectively. The elimination efficiency slightly dropped from the third cycle and the fifth cycle degraded

around 90% of hazards except for diclofenac. Most of the active sites of the  $\text{FeS}_2\text{@GO}$  composite come in contact with tested contaminants in its first cycle reaction, and the number of active sites was subsequently reduced. As a result, the removal efficiency of MPs would be decreased. Therefore, the results confirmed that the synthesized  $\text{FeS}_2\text{@GO}$  composite has good recyclability and stability.

A highly efficient and stable heterogeneous catalyst is always desired for photo-Fenton reactions. In summary, the current study reports a facile and efficient synthetic procedure for the synthesis of nano-pyrite-based  $\text{FeS}_2\text{@GO}$  composite. The uniform GO-coated layer on  $\text{FeS}_2$  can prevent the permeation of  $\text{O}_2$  and  $\text{H}_2\text{O}$ , thus promoting its photocatalytic ability. Our results confirm that the  $\text{FeS}_2\text{@GO}$ -based heterogeneous photo-Fenton process could simultaneously remove ARB, e-ARGs, and MPs from various water matrixes, even in real wastewater effluent. Such efficient performance is due to the generation of  $\text{HO}^\bullet$ ,  $^1\text{O}_2$ , and  $\text{O}_2^{\bullet-}$  by the photocatalytic action of  $\text{FeS}_2\text{@GO}$  with  $\text{H}_2\text{O}_2$ . In addition, the  $\text{FeS}_2\text{@GO}$  catalyst shows good stability and reusability in a wide pH range. It should be emphasized that the nano  $\text{FeS}_2\text{@GO}$  can be recovered by centrifugation. Further research would need to improve the  $\text{FeS}_2\text{@GO}$  catalyst separation property.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.2c03334>.

Discussions of chemicals used, structure characterization, preparation of *E. coli* and enumeration of ARB, bacterial regrowth and recovery analysis, DNA extraction, RT-PCR analysis of extracellular and intracellular ARG, analytical methods, reactive species analysis, and electron paramagnetic resonance spectroscopy, figures of standard curves of *tetA* and *blaTEM-1* genes in RT-PCR measurement, STEM, TEM, and HRTEM images, EDS elemental mappings images, inactivation of ARB in synthetic wastewater matrix, and effects of scavengers TEA, IPA, SA,  $\text{K}_2\text{Cr}_2\text{O}_7$ , SOD, and MeOH on the degradation of MP in UPW under the heterogenous photo-Fenton process, and tables of primers used for the target ARG amplicon analysis, control experiments for inactivation of ARB, and characteristics of wastewater effluent (PDF)

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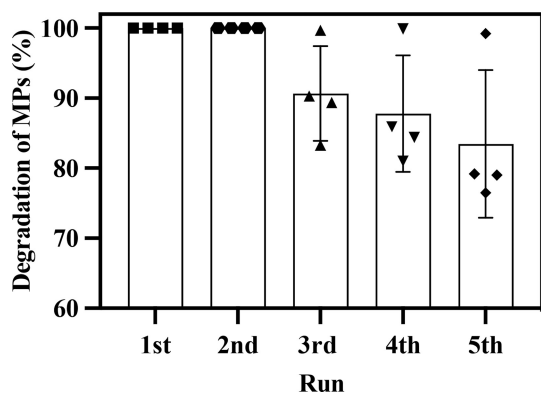
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**Figure 8.** Recycling ability of  $\text{FeS}_2\text{@GO}$  for the degradation of MPs at the doses of 0.25 g/L  $\text{FeS}_2\text{@GO}$  (8:2), 1 mM  $\text{H}_2\text{O}_2$  (34.2 mg/L), and 130  $\text{mW}/\text{cm}^2$  of light intensity.

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## Notes

The authors declare no competing financial interest.

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